

EV DISCUSSIONS ON BLUESKY

Text mining, engagement testing, clustering and reply-network analysis

COMP3020 • GROUP 9 (Ishan , thevindu , sachini)

How do people discuss electric vehicles, which topics attract likes, and how are reply conversations connected?

3,976
usable posts

1,240
“charg” mentions

p = 0.0015
category–like association

1,748
accounts in reply network

DATA & APPROACH Bluesky posts collected in R using four searches: “electric vehicles”, “electric cars”, “Tesla” and “EV charging”. After cleaning, 3,976 usable posts remained. Analyses: word frequency/word cloud; chi-square test of independence; K-means clustering on 463 posts; and directed reply-network centrality.

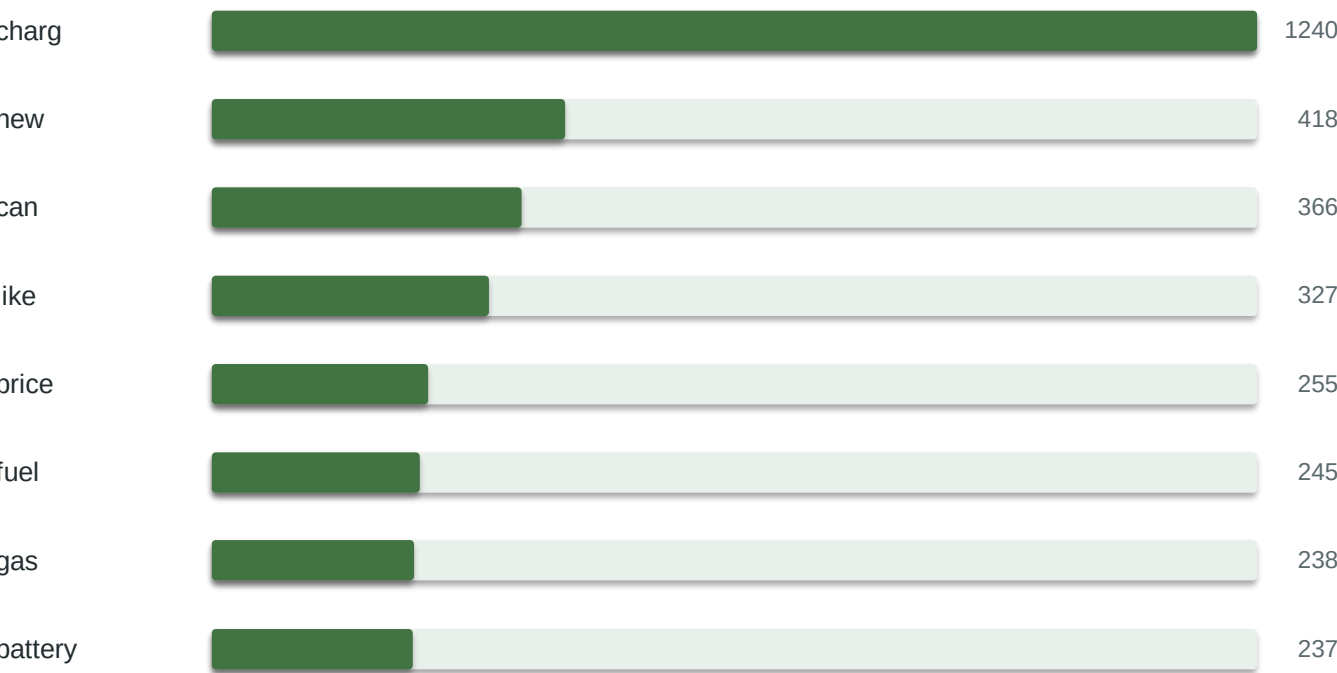
1 WHAT ARE PEOPLE DISCUSSING?

The most frequent topic-related term was “charg” (1,240 occurrences). Other recurring terms included price (255), fuel (245), gas (238), battery (237) and drive (230).

INTERPRETATION

Charging infrastructure is prominent, alongside affordability, energy use and practical EV ownership. Word frequency does not reveal whether posts support or oppose EVs.

MOST FREQUENT TERMS



2 WHAT ATTRACTS LIKES?

A chi-square test found a statistically significant association between search category and whether likes were above the median ($\chi^2 = 15.41$, $df = 3$, $p = 0.001498$).

Median likes = 1; above-median means more than 1 like. This is an association, not evidence that the search topic causes engagement.

SHARE WITH ABOVE-MEDIAN LIKES



3 CAN POSTS BE GROUPED?

K-means grouped 463 posts into three exploratory clusters. The elbow plot did not show a clear optimum, and the MDS projection showed overlap; therefore, clusters are approximate rather than sharply separated.

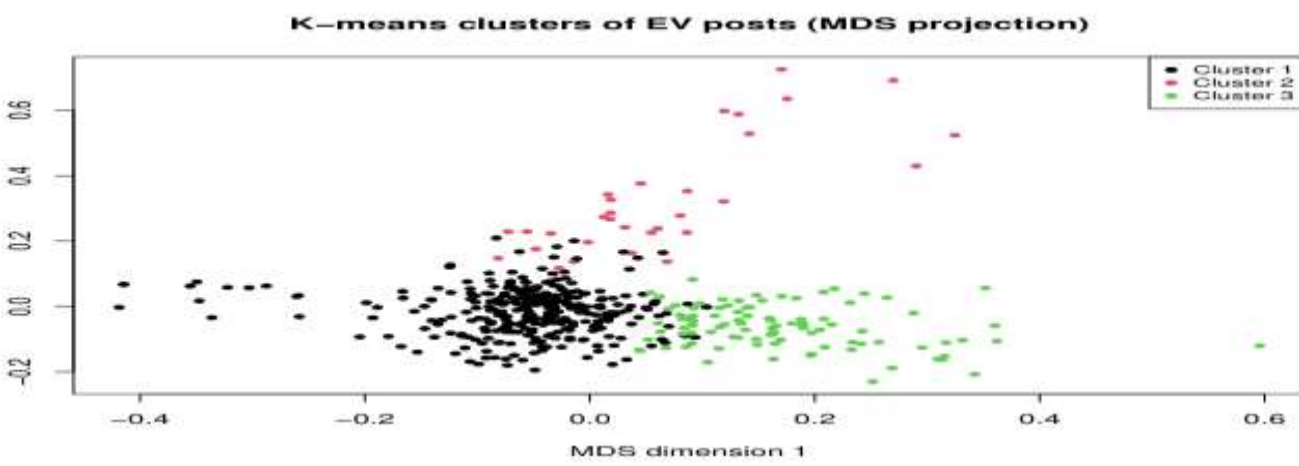
CLUSTER SIZES (n = 463)



Main themes from keywords:

- Everyday EV opinions and buying
- Market news and manufacturers
- Charging stations, home charging and cost

K-MEANS CLUSTERS: MDS PROJECTION



463 posts; clusters overlap, so themes are exploratory rather than fully separated. Source: Group 9 report, p. 8.

4 HOW ARE REPLIES CONNECTED?

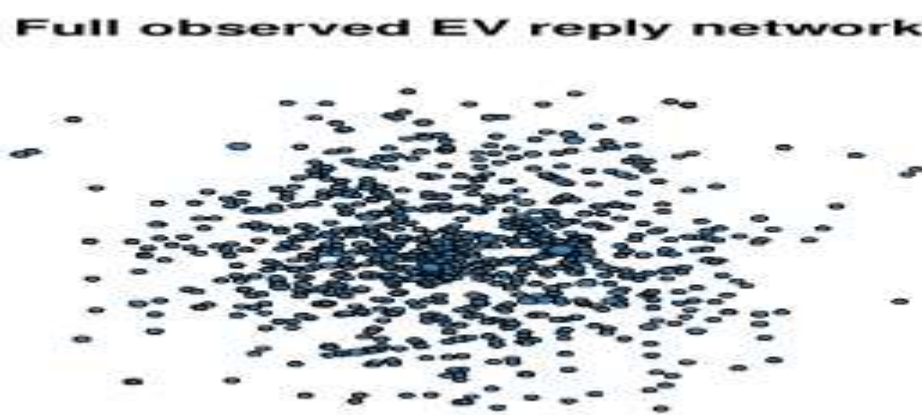
The observed reply network contained 1,748 accounts and 1,141 directed reply ties. Its low density (0.000374) and 636 weakly connected components indicate highly fragmented conversations.

NETWORK SNAPSHOT

Accounts / nodes	1,748
Reply ties / edges	1,141
Network density	0.000374
Weak components	636
Largest component	50 nodes · 52 edges

Top observed account: in-degree = 21; PageRank = 0.00782. A separate account had the highest reported betweenness (53), suggesting a possible bridging role.

OBSERVED EV REPLY NETWORK



The sparse, dispersed structure reflects fragmented sampled conversations. Source: Group 9 report, p. 11.

OVERALL CONCLUSION

Bluesky EV conversations prominently feature charging, cost and ownership practicality. Like engagement differs significantly by search category, with “Electric Cars” posts most likely to exceed the median likes (37.9%). Clustering reveals broad but overlapping themes, while reply-network analysis shows that interactions are dispersed across many disconnected conversations.

LIMITATIONS

- Four search terms and a specific collection period may omit relevant conversations.
- Word frequency does not measure sentiment.
- Clustering used only 463 posts and had no clear elbow.
- Network analysis included replies only; findings are not generalisable to all Bluesky users.

NEXT STEPS

Expand the sample and collection period; add sentiment analysis; and track how EV topics and reply networks evolve over time.